

Cognizant Digital Nurture 5.0

Deep Skilling Program

Week 7 – DevOps & Cloud

Cloud Fundamentals

Prepared By

Ashreen Fathima

B.E. Computer Science and Design
RMK Engineering College

Email: ashreenfathima26@gmail.com

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1. Introduction

Cloud computing has transformed the way organizations build, deploy, and manage applications by providing on-demand access to computing resources over the internet. Instead of purchasing and

maintaining physical infrastructure, organizations can access servers, storage, databases, networking, and software whenever required.

Cloud computing supports digital transformation by improving scalability, reducing operational costs, increasing system availability, and accelerating application deployment. It is widely used in modern software development, DevOps, artificial intelligence, data analytics, and enterprise solutions.

2. What is Cloud Computing?

Cloud computing is the delivery of computing services over the internet, allowing users to access infrastructure, platforms, and software on demand without owning physical hardware.

Definition

Cloud computing is a technology that enables users to access shared computing resources such as servers, storage, databases, networking, and applications over the internet on a pay-as-you-use basis.

3. Why Cloud Computing?

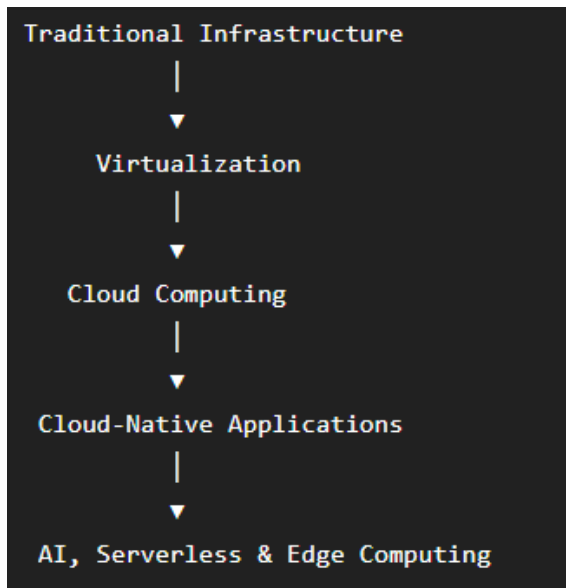
Cloud computing helps organizations improve efficiency, reduce infrastructure costs, and deliver services quickly.

Key Benefits

- On-demand resource availability
 - Cost optimization
 - High scalability
 - Faster deployment
 - Global accessibility
 - Business continuity
 - Disaster recovery support
 - Improved collaboration
-

4. Evolution of Cloud Computing

The evolution of cloud computing has changed how IT infrastructure is delivered.



5. Essential Characteristics of Cloud Computing

According to the National Institute of Standards and Technology (NIST), cloud computing has five essential characteristics.

On-Demand Self-Service

Users can provision computing resources whenever needed without requiring manual interaction with the service provider.

Broad Network Access

Cloud services are available over the internet and can be accessed from various devices such as laptops, smartphones, and tablets.

Resource Pooling

Computing resources are shared among multiple users while maintaining data isolation and security.

Rapid Elasticity

Resources can be scaled up or down automatically depending on business requirements.

Measured Service

Cloud usage is monitored, measured, and billed based on actual consumption.

6. Cloud Service Models

Cloud services are categorized into three primary models.

Infrastructure as a Service (IaaS)

Provides virtualized computing resources including servers, storage, and networking.

Examples

- Amazon EC2
 - Microsoft Azure Virtual Machines
 - Google Compute Engine
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Platform as a Service (PaaS)

Provides a platform for developing, testing, and deploying applications without managing infrastructure.

Examples

- Azure App Service
 - Google App Engine
 - Heroku
-

Software as a Service (SaaS)

Provides complete software applications through the internet.

Examples

- Microsoft 365
 - Google Workspace
 - Salesforce
-

Comparison of Service Models

Feature	IaaS	PaaS	SaaS
Infrastructure Management	User	Provider	Provider
Operating System	User	Provider	Provider
Application Management	User	User	Provider
Primary Users	Administrators	Developers	End Users

7. Cloud Deployment Models

Cloud deployment models define how cloud infrastructure is implemented.

Public Cloud

Infrastructure is owned and managed by third-party providers and shared among multiple organizations.

Examples:

- Amazon Web Services
 - Microsoft Azure
 - Google Cloud Platform
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Private Cloud

Infrastructure is dedicated to a single organization, offering greater control, customization, and security.

Hybrid Cloud

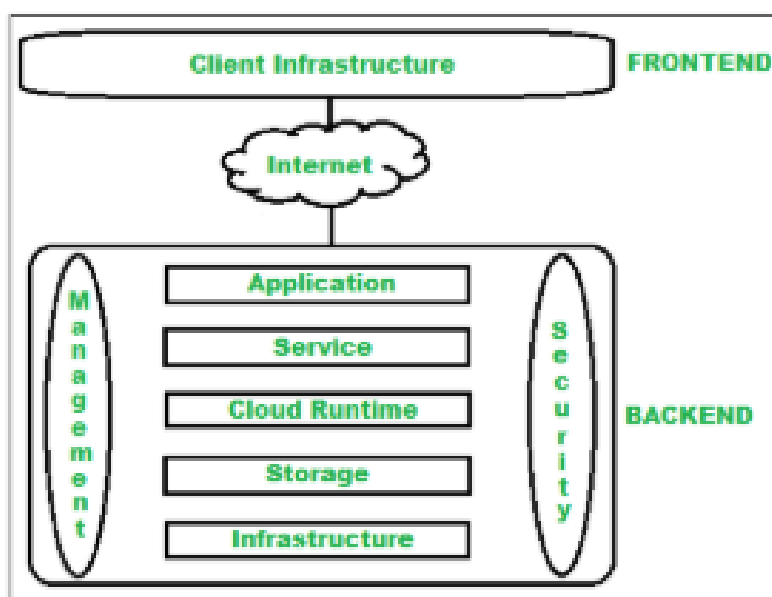
Combines public and private cloud environments, enabling organizations to move workloads between them.

Community Cloud

Infrastructure is shared by organizations with similar security, compliance, or operational requirements.

8. Cloud Architecture

Cloud architecture consists of front-end components used by clients and back-end infrastructure managed by cloud providers.



9. Major Cloud Service Providers

Several organizations provide cloud services globally.

Provider	Popular Services
Amazon Web Services (AWS)	EC2, S3, RDS, Lambda
Microsoft Azure	Virtual Machines, Azure SQL, App Services
Google Cloud Platform (GCP)	Compute Engine, Cloud Storage, BigQuery
Oracle Cloud Infrastructure (OCI)	Compute, Autonomous Database
IBM Cloud	AI, Kubernetes, Virtual Servers

10. Cloud Security

Cloud security protects cloud-based systems, applications, and data from cyber threats.

Security Practices

- Identity and Access Management (IAM)
 - Multi-Factor Authentication (MFA)
 - Data Encryption
 - Secure Network Configuration
 - Continuous Monitoring
 - Backup and Disaster Recovery
 - Compliance Management
-

11. Advantages of Cloud Computing

Cloud computing offers several advantages.

- Reduced infrastructure costs
- High scalability
- Flexible resource allocation
- Faster application deployment
- Improved collaboration
- Global accessibility
- Automatic updates
- High availability

- Disaster recovery capabilities
 - Enhanced business continuity
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12. Challenges of Cloud Computing

Although cloud computing provides many benefits, organizations should consider the following challenges.

- Internet dependency
 - Data privacy concerns
 - Vendor lock-in
 - Regulatory compliance
 - Downtime risks
 - Security misconfigurations
 - Migration complexity
-

13. Real-World Applications

Cloud computing is widely used in:

- Web application hosting
 - Mobile application development
 - Artificial Intelligence
 - Machine Learning
 - Big Data Analytics
 - Internet of Things (IoT)
 - Online education platforms
 - Video streaming services
 - Enterprise Resource Planning (ERP)
 - Disaster Recovery Systems
-

14. Best Practices

To maximize the benefits of cloud computing:

- Select the appropriate cloud service model.
- Implement strong authentication mechanisms.

- Encrypt sensitive information.
 - Monitor cloud resource utilization regularly.
 - Perform scheduled backups.
 - Follow organizational security policies.
 - Optimize resource usage to reduce costs.
 - Ensure compliance with industry standards.
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15. Conclusion

Cloud computing has become an essential component of modern information technology by providing scalable, secure, and cost-effective computing resources over the internet. It enables organizations to innovate rapidly, improve operational efficiency, and support digital transformation initiatives. Understanding cloud computing concepts forms a strong foundation for careers in software engineering, DevOps, cloud architecture, and enterprise application development.